Halima Abdirizak Mohamed

Security-Focused Software Engineer

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SUMMARY

Informatics undergraduate with internship experience in backend engineering and cybersecurity. Experienced in API development, application security, and software testing, with an interest in building reliable, secure systems. Seeking a backend or software engineering internship.

EDUCATION

President University, Indonesia Expected Graduation: 2027

Bachelor of Information Technology, Cybersecurity Concentration | CGPA: 3.73/4.00

Relevant Coursework: Web Development, Network Security, Cryptography, Risk Management

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, PHP, HTML/CSS

Backend & Web: FastAPI, Flask, Node.js, Express, React, Next.js, REST APIs, JWT / RBAC Authentication

Databases & AI: PostgreSQL, MongoDB, SQLite, SQLAlchemy, Ollama (Local LLM Integration)

Tools & Testing: Git, GitHub, Docker, Linux (Kali), pytest, Automated & Regression Testing

Cybersecurity: Web Application Security, Vulnerability Detection, Phishing Analysis, NIST CSF, OWASP Top 10, SafeLine WAF

EXPERIENCE

ElectraWireless Sep 2026 – Present

Cybersecurity Intern *Remote*

- Developed a **SHA-256 hash-comparison malware detection** module in Python for an IoT firmware vulnerability scanner, integrated into the existing firmware extraction pipeline.
- Wrote automated tests for known malware hashes, unknown files, and a seeded benign signature; converted a network-scanner script into a **pytest suite** and fixed an IPv6 parsing bug affecting live scans.
- Maintained a regression suite of **130+ tests** across successive integrations in a multi-contributor Git workflow.

FlyRank AI Jul 2026 – Sep 2026

Backend AI Engineering Intern (Task-Based) *Remote*

- Built Midtrans payment-webhook handling for the FlyRank Billing Engine (FastAPI) with **HMAC-SHA512 signature verification and idempotency controls**, reducing unauthorized and duplicate payment events.
- Designed usage metering, subscription quotas, and **tenant data isolation** for Free and Pro plans using FastAPI, PostgreSQL, SQLAlchemy, and Alembic; validated by **68 automated tests**.

PROJECTS

AuditShield – NIST CSF 2.0 Cybersecurity Audit Platform *Flask, SQLite, Ollama*

- Built an audit platform that maps **51 NIST CSF controls** to an evidence-driven workflow: asset inventory, risk scoring (likelihood × impact), findings, and exportable compliance reports.
- Added role-based access (Admin / Auditor / Auditee), immutable evidence collection, and a **local LLM assistant (Ollama)** that explains vulnerabilities in plain language.

SEMIS – Secure Employee Management Information System *Node.js, Express, React, MongoDB*

- Built a security-first HR platform with **MFA (email OTP)**, 15-minute **JWT** sessions, bcrypt hashing, RBAC middleware, and **AES-256 encryption** for sensitive fields such as salary and national ID.
- Hardened it against OWASP Top 10 threats with rate limiting, 5-attempt account lockout, CSRF tokens, security headers, input validation, and append-only audit logging.

Hookline – Phishing Detection Chrome Extension *TypeScript, React, FastAPI*

- Built a Gmail extension that flags phishing through typosquatting (~24 brands), homoglyphs, sender / reply-to mismatch, and urgency language, showing **human-readable evidence** instead of a raw score.
- Benchmarked on a **50k-URL dataset** (76.1% precision) with a reproducible evaluation harness and unit tests; added an optional **FastAPI** backend for WHOIS, DNS, and SSL enrichment.

WAF HomeLab – SafeLine Web Application Firewall *SafeLine, Docker, DVWA, Kali*

- Built an attack-and-defend lab (Kali attacker, DVWA behind SafeLine) and tested it: **SQL injection and reflected XSS blocked**, HTTP floods rate-limited, command injection detected and logged but not blocked.

CERTIFICATIONS

Google Cybersecurity Professional Certificate | GitHub Foundations (GH-900) | AWS Security Fundamentals